KELT-16b

R-1123

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_KELT-16_210721` · created 2026-07-30T22:13:49+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459416.817 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.014
Transit depth (Rp/Rs)^2	1.75 +/- 0.37 [%]
Orbital Inclination (inc)	84.21 +/- 2.9
Airmass coefficient 1 (a1)	1.085 +/- 0.019
Airmass coefficient 2 (a2)	-0.076 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	#1 - [376, 382]
Optimal Aperture	3.8
Optimal Annulus	10.93
Transit Duration (day)	0.1047 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.132 ± 0.014 vs 0.107 (deviation 1.31σ)
Duration fit vs published	0.1047 d vs 0.1038 d (deviation 0.13σ)
Mid-time O–C	-9.1 min
Depth SNR	6.9
Residual scatter	1.77 %

Light curve

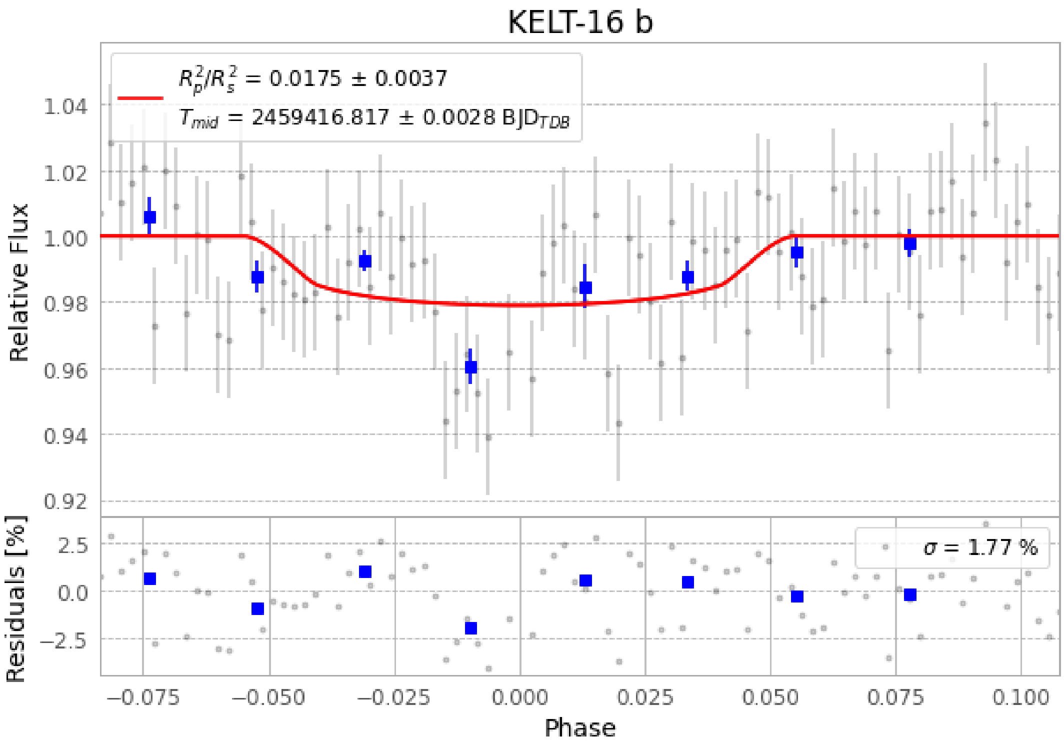

Figure 1 — FinalLightCurve_KELT-16 b_2021-07-20.png (SHA-256 482d3ee8e33635c9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 200], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f8b1dca320ce355d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

90 FITS frames (59 MB), `KELT-16210721053312.FITS` → `KELT-16210721100012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-16-210721.log (dc7dd153e6a3…), FinalLightCurve_KELT-16 b_2021-07-20.png (482d3ee8e336…), FinalLightCurve_KELT-16 b_2021-07-20.csv (37ccee541305…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 22:13Z.